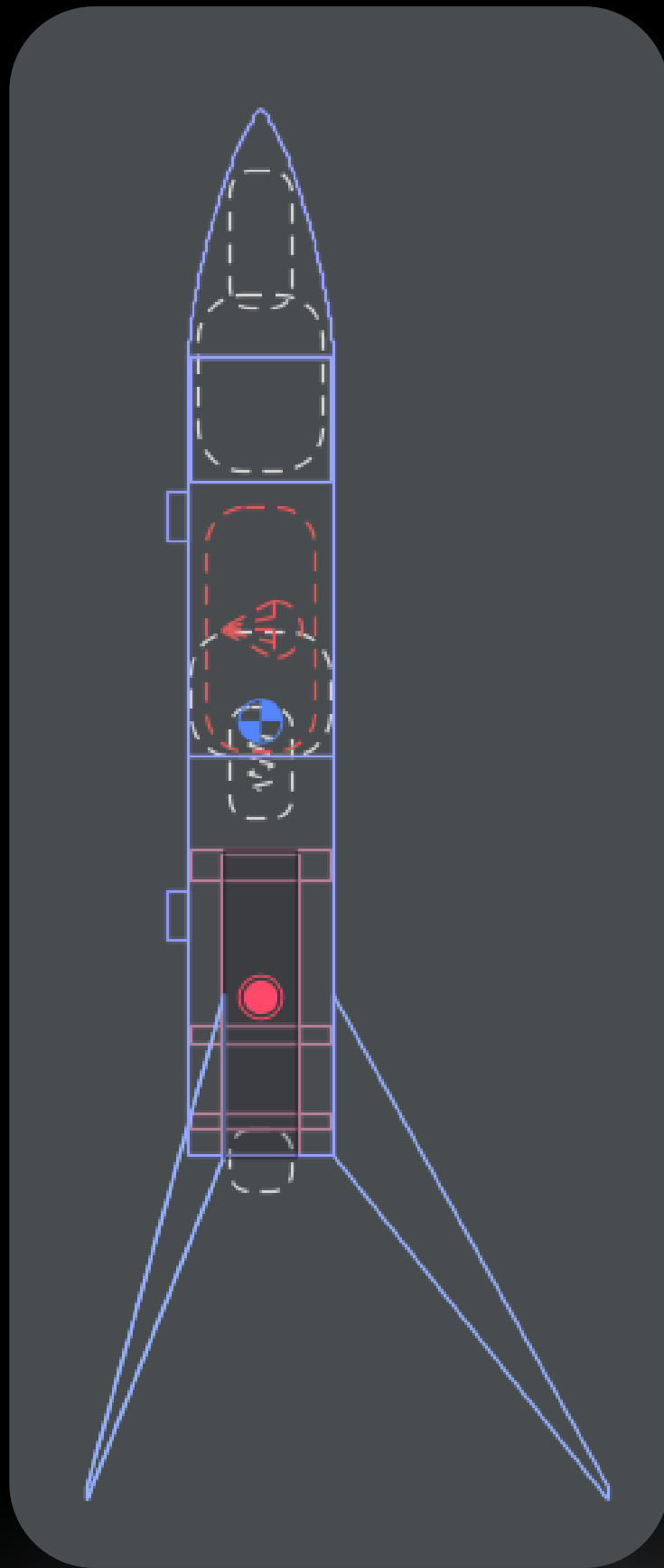




REQUIREMENTS

- Apogee: target = 3000ft
- Stability: 1.5-2.5 cal
- Off the rail speed: $>75\text{ft/s}$
- Send and recover egg without breaking
- Time to apogee: $<15.5\text{ sec}$
- Collect data on apogee, inertia, and stability during flight
- Two launch lugs

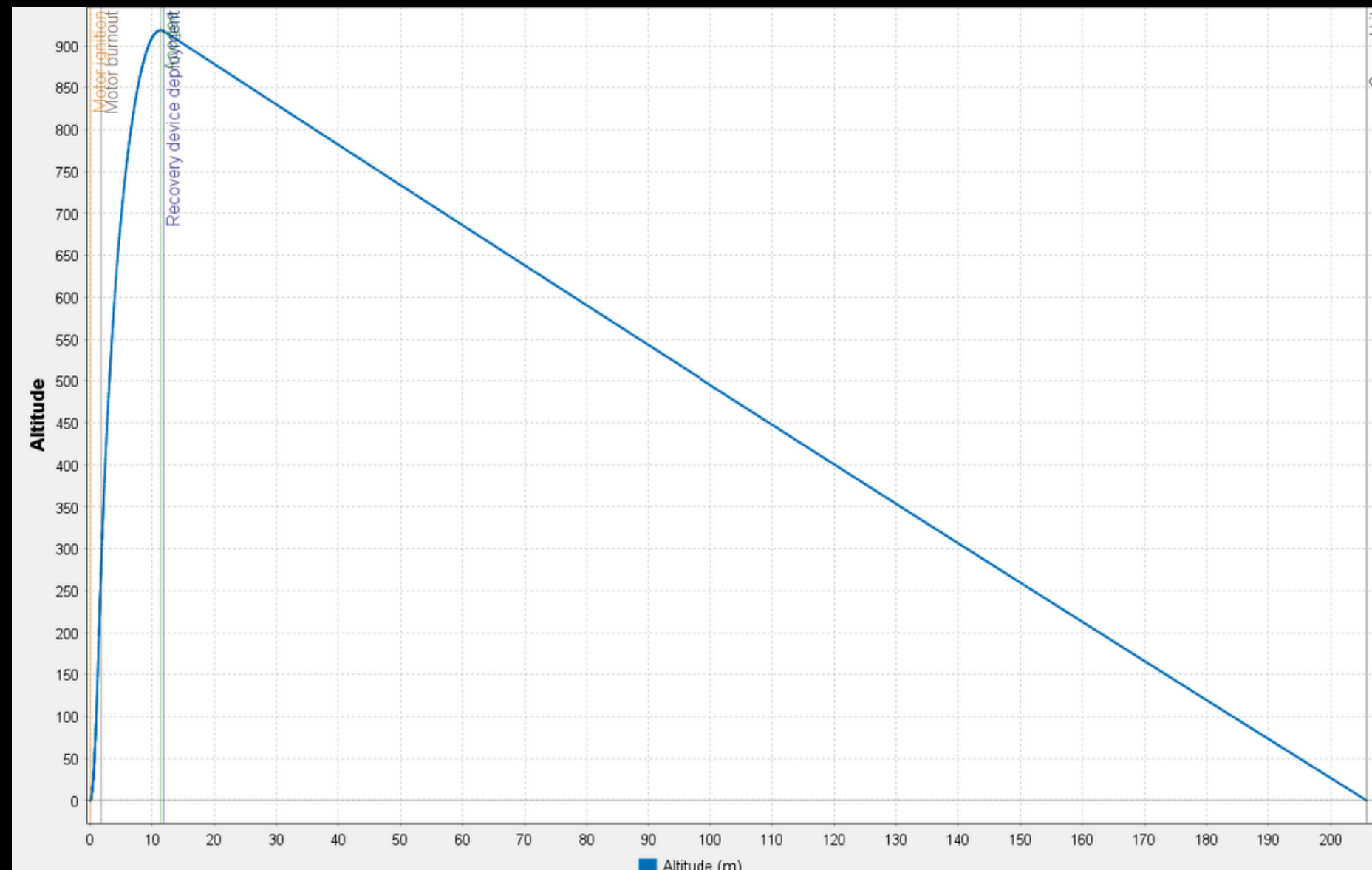


- Nose Cone:
 - Von Karman
 - contains egg, electronics
 - material: PLA (3d print)
- 1 body tube: OD = 5.82cm
 - material: fiberglass
- 3 fins; trapezoidal
 - material: $\frac{1}{8}$ " plywood
- Interior: inner tube for motor,
 - 2 centering rings
 - bulkhead
 - material $\frac{1}{2}$ " plywood

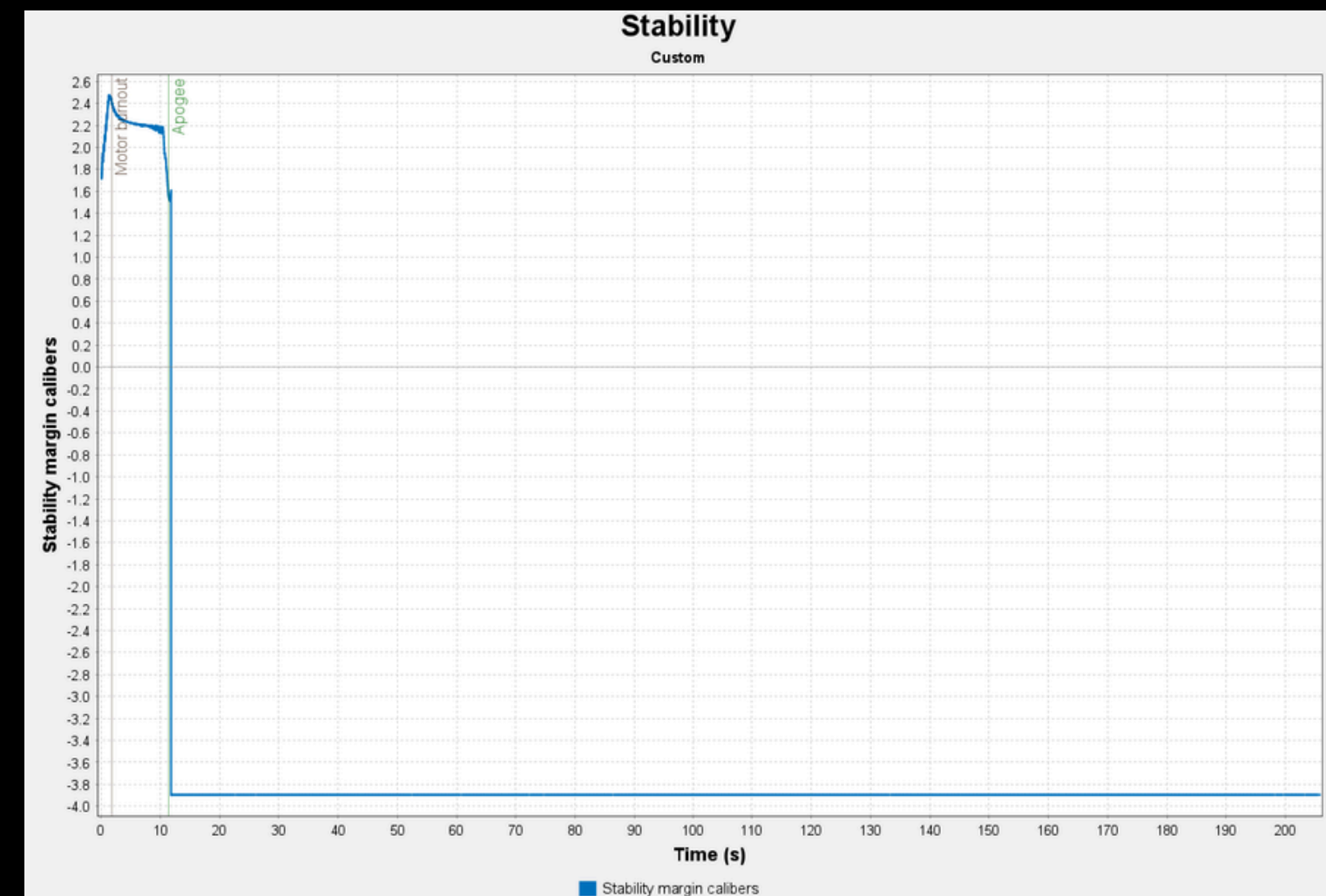
- Mass: 436g with motor
- 307g without motor

THE DESIGN

SIM STATS



- Velocity off rail: 28m/s
 - 6ft rail
- Apogee: 919m (3015ft)
- Stability: 1.91cal
- Time to apogee: 11.3 sec
- Velocity at deployment: -5 m/s
- Stability at apogee: 1.6
- Max velocity: 237m/s
- Max mach #: 0.7
- Total Impulse: 133Ns
- Max thrust: 102N
- Avg thrust: 78N
- Descent time: 194 sec



AIRFRAME



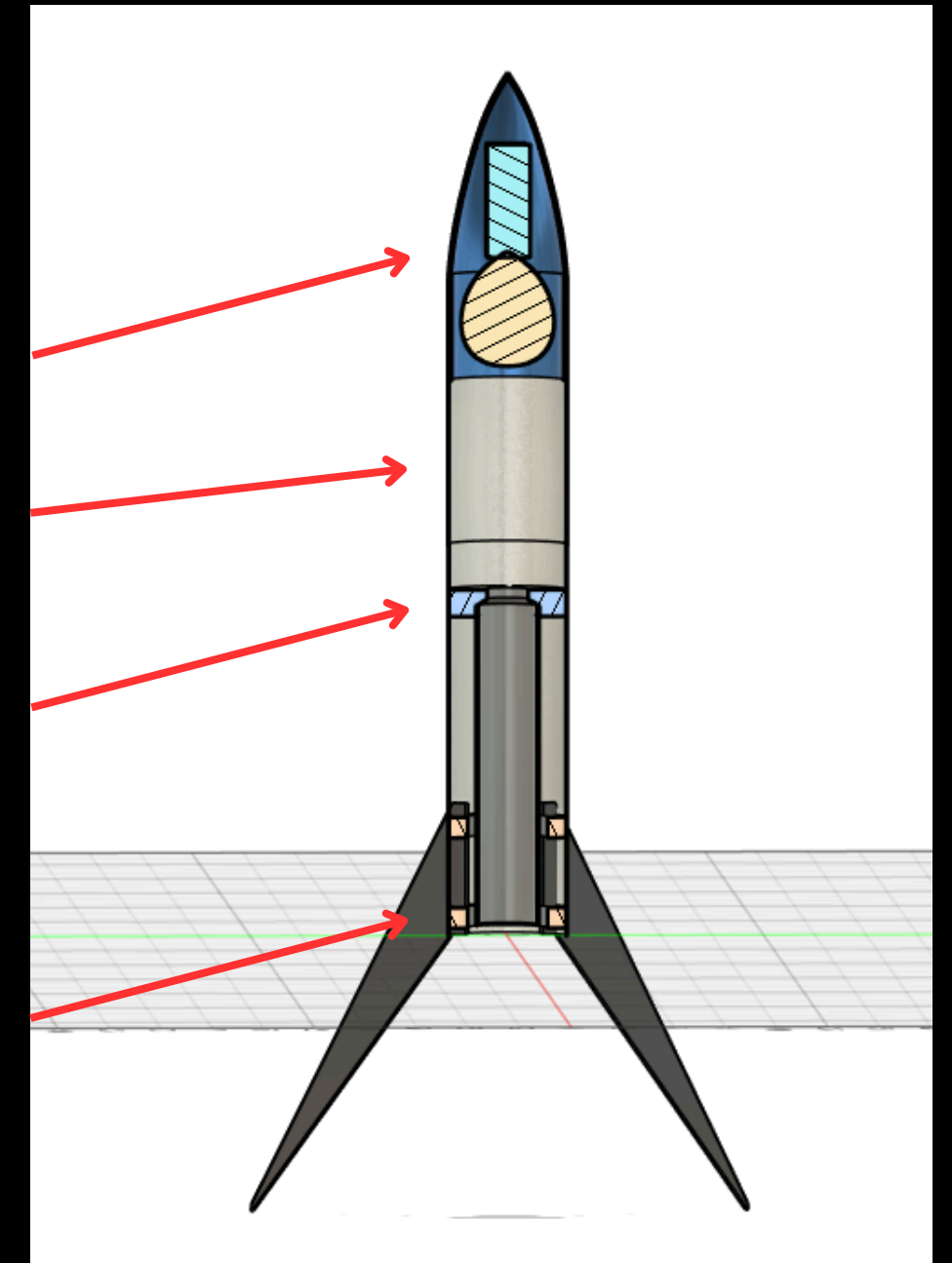
- Fiberglass body tube
- ID = 5.72cm (2.25")
- OD = 5.82cm
 - thickness = 0.05cm
- length = 32cm

INTERNAL COMPONENTS



- parachute:
 - 28g
 - 30" diameter
- flame blanket:
 - 9" side length (square)
 - 10g
- Shock cord:
 - 2ft length
 - nylon-covered latex
 - 0.4cm diameter

- Egg and electronics in nose cone
- Parachute, flame blanket, and shock cord will go here
- Bulkhead at end of inner tube with hole for ejection fuse
- 2 centering rings within fin root chord length



PAYLOAD SAFETY

